

31.3 Singular covariance matrices

A covariance matrix is nonnegative definite, not necessarily positive definite. When it is singular, some linear combination of the observations has zero variance and is therefore a *constant*: the data satisfy an exact linear relation. This is not a pathology to be regularized away — it happens whenever a linear operation has destroyed information — and it changes the theory, because some functions of β become known without error.

31.3.1 Where a singular covariance comes from

Three mechanisms cover most cases.

Constraints built into the data. Percentages adjusted to add to a hundred, effects reported under a sum-to-zero side condition (Chapter 8), residuals from an earlier fit: the reported vector satisfies an exact linear identity, and if $\mathbf{a}^\top \mathbf{Y}$ is constant then $\mathbf{V}\mathbf{a} = \mathbf{0}$ by Theorem 2.2.3(c).

A deterministic component. Some observations may be free of measurement error — a boundary value known exactly, a fitted value substituted for a missing one — and the corresponding rows and columns of $\mathbf{V}$ vanish. So is $\mathbf{V} = \mathbf{ZGZ}^\top$, the covariance of a random-effects model $\mathbf{Y} = \mathbf{X}\beta + \mathbf{Zu}$ with no separate measurement error, as soon as $\mathbf{Z}$ has fewer columns than rows; Chapter 32 studies that family.

Linear filtering with redundancy. Replacing uncorrelated observations by more linear combinations than the rank of the transformation — differences together with their total, overlapping averages together with the underlying ones — produces a singular covariance. Redundancy is what matters, not filtering: $n - 1$ successive differences of n uncorrelated observations have a nonsingular tridiagonal covariance.

31.3.2 The consistency condition

Idea. A singular $\mathbf{V}$ confines the *errors* to a subspace: $\Pr(\boldsymbol{\varepsilon} \in \mathcal{C}(\mathbf{V})) = 1$. With $E(\mathbf{Y}) = \mathbf{X}\beta \in \mathcal{C}(\mathbf{X})$, this confines the data to $\mathcal{C}(\mathbf{X} : \mathbf{V}) = \mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V})$ and $\mathbf{X}\beta$ to the affine set $\mathbf{y} + \mathcal{C}(\mathbf{V})$. Estimation becomes a constrained problem: find the point of $\mathcal{C}(\mathbf{X})$ nearest $\mathbf{y}$ among those the data allow.

Write $\mathcal{C}(\mathbf{X} : \mathbf{V})$ for $\mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V})$, the column space of $[\mathbf{X} \ \mathbf{V}]$. The following small lemma is used repeatedly.

Lemma 31.3.1

If $\mathbf{A}$ and $\mathbf{B}$ are nonnegative definite of the same order, then $\mathcal{C}(\mathbf{A} + \mathbf{B}) = \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$.

Proof. $(\mathbf{A} + \mathbf{B})\mathbf{w} = \mathbf{Aw} + \mathbf{Bw}$ shows $\mathcal{C}(\mathbf{A} + \mathbf{B}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$. For the reverse, take $\mathbf{w} \perp \mathcal{C}(\mathbf{A} + \mathbf{B})$. Then $\mathbf{w}^\top \mathbf{Aw} + \mathbf{w}^\top \mathbf{Bw} = 0$, and both terms are nonnegative, so both vanish. By Theorem 1.7.6, $\|\mathbf{A}^{1/2}\mathbf{w}\|^2 = \mathbf{w}^\top \mathbf{Aw} = 0$, so $\mathbf{Aw} = \mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{w} = \mathbf{0}$, and likewise $\mathbf{Bw} = \mathbf{0}$. Hence $\mathbf{w} \perp \mathcal{C}(\mathbf{A})$ and $\mathbf{w} \perp \mathcal{C}(\mathbf{B})$, so $\mathcal{C}(\mathbf{A} + \mathbf{B})^\perp \subseteq (\mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B}))^\perp$, which gives the inclusion.

31.3.3 The general theorem

Fix an orthogonal splitting adapted to $\mathbf{V}$: let $\mathbf{\Gamma}$ be $n \times q$ and $\mathbf{N}$ be $n \times (n - q)$ with orthonormal columns, $\mathcal{C}(\mathbf{\Gamma}) = \mathcal{C}(\mathbf{V})$ and $\mathcal{C}(\mathbf{N}) = \mathcal{C}(\mathbf{V})^\perp$, where $q = \text{rank}(\mathbf{V})$. Put $\mathbf{\Lambda} = \mathbf{\Gamma}^\top \mathbf{V} \mathbf{\Gamma}$, which is $q \times q$ positive definite, and let $\mathbf{V}^+$ be the Moore–Penrose inverse (Definition 1.3.2), so that $\mathbf{V}^+ = \mathbf{\Gamma} \mathbf{\Lambda}^{-1} \mathbf{\Gamma}^\top$. Write $r = \text{rank}(\mathbf{X})$ and $d = \dim(\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V}))$, and let $\mathbf{U}$ be any $n \times d$ matrix of full column rank with $\mathcal{C}(\mathbf{U}) = \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V})$.

Theorem 31.3.2. The general Gauss–Markov model

Assume Eq. 31.1.1 with $\mathbf{V}$ nonnegative definite of rank q .

- (a) (*Consistency.*) $\mathbf{N}^\top \mathbf{Y} = \mathbf{N}^\top \mathbf{X} \boldsymbol{\beta}$ with probability one; consequently $\Pr(\mathbf{Y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})) = 1$ and $\Pr(\mathbf{X} \boldsymbol{\beta} \in \mathbf{Y} + \mathcal{C}(\mathbf{V})) = 1$.
- (b) (*Estimability and perfect estimation.*) $\lambda^\top \boldsymbol{\beta}$ has a linear unbiased estimator iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, exactly as when $\mathbf{V} = \mathbf{I}$. It has a linear unbiased estimator of variance zero iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top \mathbf{N})$, a subspace of dimension $r - d$, and then $\lambda^\top \boldsymbol{\beta} = \mathbf{b}^\top \mathbf{N}^\top \mathbf{Y}$ with probability one for any $\mathbf{b}$ with $\mathbf{X}^\top \mathbf{N} \mathbf{b} = \boldsymbol{\lambda}$.
- (c) (*Rao’s unified formula.*) Let $\mathbf{T} = \mathbf{V} + \mathbf{X} \mathbf{X}^\top$. Then $\mathcal{C}(\mathbf{T}) = \mathcal{C}(\mathbf{X} : \mathbf{V})$, $\mathbf{T} \mathbf{T}^+ \mathbf{X} = \mathbf{X}$, and

$$\mathbf{A} = \mathbf{X}(\mathbf{X}^\top \mathbf{T}^+ \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{T}^+ \quad (31.3.1)$$

satisfies $\mathbf{A} \mathbf{X} = \mathbf{X}$ and $\mathbf{A} \mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$. Hence $\mathbf{A} \mathbf{Y}$ is a BLUE of $\mathbf{X} \boldsymbol{\beta}$.

- (d) (*Uniqueness.*) Every estimable $\lambda^\top \boldsymbol{\beta}$ has a BLUE, and any two BLUEs of it agree for every $\mathbf{y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})$, hence with probability one.
- (e) (*Projection form.*) Let $\mathbf{y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})$. The set $\mathcal{F}(\mathbf{y}) = \{\mathbf{b} : \mathbf{y} - \mathbf{X} \mathbf{b} \in \mathcal{C}(\mathbf{V})\}$ is nonempty, and

$$\tilde{\boldsymbol{\beta}} \in \arg \min_{\mathbf{b} \in \mathcal{F}(\mathbf{y})} (\mathbf{y} - \mathbf{X} \mathbf{b})^\top \mathbf{V}^+ (\mathbf{y} - \mathbf{X} \mathbf{b}) \quad (31.3.2)$$

determines $\mathbf{X} \tilde{\boldsymbol{\beta}}$ uniquely; the map $\mathbf{y} \mapsto \mathbf{X} \tilde{\boldsymbol{\beta}}$ is linear and $\mathbf{X} \tilde{\boldsymbol{\beta}}$ is a BLUE of $\mathbf{X} \boldsymbol{\beta}$. Explicitly, with $\mathbf{K} = \mathbf{N}^\top \mathbf{X}$ and any generalized inverse $\mathbf{K}^-$,

$$\mathbf{X} \tilde{\boldsymbol{\beta}} = \mathbf{X} \mathbf{b}_* + \mathbf{U}(\mathbf{U}^\top \mathbf{V}^+ \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{V}^+ (\mathbf{y} - \mathbf{X} \mathbf{b}_*), \quad \mathbf{b}_* = \mathbf{K}^- \mathbf{N}^\top \mathbf{y}. \quad (31.3.3)$$

- (f) (*Error sum of squares.*) Put $\hat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \mathbf{X} \tilde{\boldsymbol{\beta}}$ and $\text{SSE}_{\mathbf{V}} = \hat{\boldsymbol{\varepsilon}}^\top \mathbf{V}^+ \hat{\boldsymbol{\varepsilon}}$. Then $E(\text{SSE}_{\mathbf{V}}) = \sigma^2(q - d)$, and $q - d = \text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$. If $\mathbf{Y}$ is normal, then $\text{SSE}_{\mathbf{V}}/\sigma^2 \sim \chi^2(q - d)$, independently of $\mathbf{X} \tilde{\boldsymbol{\beta}}$.

Proof.

- (a) By Theorem 2.2.3(d), $\Pr(\boldsymbol{\varepsilon} \in \mathcal{C}(\sigma^2 \mathbf{V})) = 1$, and $\mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{N})^\perp$, so $\mathbf{N}^\top \boldsymbol{\varepsilon} = \mathbf{0}$ almost surely, giving $\mathbf{N}^\top \mathbf{Y} = \mathbf{N}^\top \mathbf{X} \boldsymbol{\beta}$. Then $\mathbf{Y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon} \in \mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V})$ and $\mathbf{X} \boldsymbol{\beta} = \mathbf{Y} - \boldsymbol{\varepsilon} \in \mathbf{Y} + \mathcal{C}(\mathbf{V})$.
- (b) $\mathbf{c}^\top \mathbf{Y}$ is unbiased for $\lambda^\top \boldsymbol{\beta}$ for every $\boldsymbol{\beta}$ iff $\mathbf{X}^\top \mathbf{c} = \boldsymbol{\lambda}$, which is possible iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$. Its variance is $\sigma^2 \mathbf{c}^\top \mathbf{V} \mathbf{c}$, which vanishes iff $\mathbf{V} \mathbf{c} = \mathbf{0}$ (as in Lemma 31.3.1), that is iff

$\mathbf{c} \in \mathcal{C}(\mathbf{V})^\perp = \mathcal{C}(\mathbf{N})$, say $\mathbf{c} = \mathbf{N}\mathbf{b}$. Then $\boldsymbol{\lambda} = \mathbf{X}^\top \mathbf{N}\mathbf{b}$ and $\mathbf{c}^\top \mathbf{Y} = \mathbf{b}^\top \mathbf{N}^\top \mathbf{Y}$, which equals $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ almost surely by (a). For the dimension, $\text{rank}(\mathbf{X}^\top \mathbf{N}) = \text{rank}(\mathbf{N}) - \dim\{\mathbf{b} : \mathbf{N}\mathbf{b} \perp \mathcal{C}(\mathbf{X})\}$ by Theorem 1.2.2, and $\{\mathbf{N}\mathbf{b} : \mathbf{N}\mathbf{b} \perp \mathcal{C}(\mathbf{X})\} = \mathcal{C}(\mathbf{V})^\perp \cap \mathcal{C}(\mathbf{X})^\perp = (\mathcal{C}(\mathbf{X} : \mathbf{V}))^\perp$, of dimension $n - \text{rank}(\mathbf{X} : \mathbf{V})$. Since $\text{rank}(\mathbf{X} : \mathbf{V}) = r + q - d$, we get $\text{rank}(\mathbf{X}^\top \mathbf{N}) = (n - q) - (n - r - q + d) = r - d$.

- (c) By Lemma 31.3.1 and $\mathcal{C}(\mathbf{X}\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})$ (Corollary 6.2.4), $\mathcal{C}(\mathbf{T}) = \mathcal{C}(\mathbf{V}) + \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X} : \mathbf{V})$. Since $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{T})$, the system $\mathbf{T}\mathbf{H} = \mathbf{X}$ is consistent (Theorem 1.4.1), so $\mathbf{X} = \mathbf{T}\mathbf{H}$ for some $\mathbf{H}$ and $\mathbf{T}\mathbf{T}^+ \mathbf{X} = \mathbf{T}\mathbf{T}^+ \mathbf{T}\mathbf{H} = \mathbf{T}\mathbf{H} = \mathbf{X}$.

Write $\mathbf{T}^+ = \mathbf{L}\mathbf{L}^\top$, possible because $\mathbf{T}^+$ is symmetric nonnegative definite, and set $\mathbf{X}_* = \mathbf{L}^\top \mathbf{X}$, so that $\mathbf{X}^\top \mathbf{T}^+ \mathbf{X} = \mathbf{X}_*^\top \mathbf{X}_*$. If $\mathbf{X}_* \mathbf{w} = \mathbf{0}$ then $\mathbf{X}\mathbf{w} \in \mathcal{N}(\mathbf{L}^\top) = \mathcal{N}(\mathbf{T}^+) = \mathcal{C}(\mathbf{T})^\perp$, while $\mathbf{X}\mathbf{w} \in \mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{T})$; hence $\mathbf{X}\mathbf{w} = \mathbf{0}$. Every column of $\mathbf{I} - (\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{X}_*$ lies in $\mathcal{N}(\mathbf{X}_*^\top \mathbf{X}_*) = \mathcal{N}(\mathbf{X}_*)$, so $\mathbf{X}(\mathbf{I} - (\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{X}_*) = \mathbf{0}$, which is $\mathbf{A}\mathbf{X} = \mathbf{X}(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{L}^\top \mathbf{X} = \mathbf{X}(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{X}_* = \mathbf{X}$.

For the second condition, let $\mathbf{d}$ satisfy $\mathbf{X}^\top \mathbf{d} = \mathbf{0}$. Using $\mathbf{V} = \mathbf{T} - \mathbf{X}\mathbf{X}^\top$ and the symmetry of $\mathbf{T}$ and $\mathbf{T}^+$,

$$\mathbf{X}^\top \mathbf{T}^+ \mathbf{V}\mathbf{d} = \mathbf{X}^\top \mathbf{T}^+ \mathbf{T}\mathbf{d} - \mathbf{X}^\top \mathbf{T}^+ \mathbf{X}\mathbf{X}^\top \mathbf{d} = (\mathbf{T}\mathbf{T}^+ \mathbf{X})^\top \mathbf{d} = \mathbf{X}^\top \mathbf{d} = \mathbf{0},$$

so $\mathbf{A}\mathbf{V}\mathbf{d} = \mathbf{0}$. As $\mathbf{d}$ ranges over $\mathcal{C}(\mathbf{X})^\perp$ this says $\mathbf{A}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$. Now Lemma 31.2.1(c) makes $\mathbf{A}\mathbf{Y}$ a BLUE of $\mathbf{X}\boldsymbol{\beta}$.

- (d) Existence is (c): for estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$, the statistic $\boldsymbol{\rho}^\top \mathbf{A}\mathbf{Y}$ is a BLUE. Uniqueness is Lemma 31.2.1(b).
- (e) If $\mathbf{y} = \mathbf{X}\mathbf{b}_0 + \mathbf{v}$ with $\mathbf{v} \in \mathcal{C}(\mathbf{V})$ then $\mathbf{b}_0 \in \mathcal{F}(\mathbf{y})$, so $\mathcal{F}(\mathbf{y}) \neq \emptyset$; and $\mathbf{b} \in \mathcal{F}(\mathbf{y})$ iff $\mathbf{K}\mathbf{b} = \mathbf{N}^\top \mathbf{y}$, a consistent system, whose solutions are $\mathbf{K}^{-1} \mathbf{N}^\top \mathbf{y}$ plus $\mathcal{N}(\mathbf{K})$. Since $\mathbf{X}\mathcal{N}(\mathbf{K}) = \{\mathbf{X}\mathbf{h} : \mathbf{N}^\top \mathbf{X}\mathbf{h} = \mathbf{0}\} = \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{U})$, the fitted vectors available are $\mathbf{X}\mathbf{b} = \mathbf{X}\mathbf{b}_* + \mathbf{U}\mathbf{c}$ with $\mathbf{b}_* = \mathbf{K}^{-1} \mathbf{N}^\top \mathbf{y}$ and $\mathbf{c} \in \mathbb{R}^d$ free. Also $\mathbf{y} - \mathbf{X}\mathbf{b}_* \in \mathcal{C}(\mathbf{V})$, and on $\mathcal{C}(\mathbf{V})$ the form $\mathbf{a}^\top \mathbf{V}^+ \mathbf{a}$ is positive definite; so minimizing over $\mathbf{c}$ is an ordinary generalized least squares problem in the $\mathbf{V}^+$ inner product with model matrix $\mathbf{U}$ of full column rank, whose unique solution is $\hat{\mathbf{c}} = (\mathbf{U}^\top \mathbf{V}^+ \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{V}^+ (\mathbf{y} - \mathbf{X}\mathbf{b}_*)$. This gives Eq. 31.3.3, shows that $\mathbf{X}\tilde{\boldsymbol{\beta}}$ is unique and that the map is linear, say $\mathbf{X}\tilde{\boldsymbol{\beta}} = \mathbf{A}_* \mathbf{y}$. It remains to check the two conditions of Lemma 31.2.1(c) for $\mathbf{A}_*$. If $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$ then $\mathbf{K}\mathbf{b}_* = \mathbf{N}^\top \mathbf{X}\boldsymbol{\beta}$ and $\mathbf{N}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}_*) = \mathbf{0}$, so $\mathbf{y} - \mathbf{X}\mathbf{b}_* \in \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{U})$; the $\mathbf{V}^+$ -projection onto $\mathcal{C}(\mathbf{U})$ leaves it fixed, and $\mathbf{A}_* \mathbf{X}\boldsymbol{\beta} = \mathbf{X}\mathbf{b}_* + (\mathbf{y} - \mathbf{X}\mathbf{b}_*) = \mathbf{X}\boldsymbol{\beta}$. Thus $\mathbf{A}_* \mathbf{X} = \mathbf{X}$. If $\mathbf{y} = \mathbf{V}\mathbf{d}$ with $\mathbf{X}^\top \mathbf{d} = \mathbf{0}$, then $\mathbf{N}^\top \mathbf{V}\mathbf{d} = \mathbf{0}$ so we may take $\mathbf{b}_* = \mathbf{0}$, and $\mathbf{U}^\top \mathbf{V}^+ \mathbf{V}\mathbf{d} = \mathbf{U}^\top \boldsymbol{\Gamma} \boldsymbol{\Gamma}^\top \mathbf{d} = \mathbf{U}^\top \mathbf{d} = \mathbf{0}$, using $\mathbf{V}^+ \mathbf{V} = \boldsymbol{\Gamma} \boldsymbol{\Gamma}^\top$, $\mathcal{C}(\mathbf{U}) \subseteq \mathcal{C}(\mathbf{V})$ and $\mathcal{C}(\mathbf{U}) \subseteq \mathcal{C}(\mathbf{X})$. So $\mathbf{A}_* \mathbf{V}\mathbf{d} = \mathbf{0}$, that is $\mathbf{A}_* \mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$.
- (f) Put $\mathbf{Z}_* = \boldsymbol{\Lambda}^{-1/2} \boldsymbol{\Gamma}^\top (\mathbf{I} - \mathbf{X}\mathbf{K}^{-1} \mathbf{N}^\top) \mathbf{Y}$, a random vector in $\mathbb{R}^q$. Because $\mathbf{V}\mathbf{N} = \mathbf{0}$, $(\mathbf{I} - \mathbf{X}\mathbf{K}^{-1} \mathbf{N}^\top) \mathbf{V}(\mathbf{I} - \mathbf{X}\mathbf{K}^{-1} \mathbf{N}^\top)^\top = \mathbf{V}$, so $\text{Cov}(\mathbf{Z}_*) = \sigma^2 \boldsymbol{\Lambda}^{-1/2} \boldsymbol{\Gamma}^\top \mathbf{V} \boldsymbol{\Gamma} \boldsymbol{\Lambda}^{-1/2} = \sigma^2 \mathbf{I}_q$. Its mean is $\boldsymbol{\Lambda}^{-1/2} \boldsymbol{\Gamma}^\top \mathbf{X}(\mathbf{I} - \mathbf{K}^{-1} \mathbf{K}) \boldsymbol{\beta}$, which lies in $\boldsymbol{\Lambda}^{-1/2} \boldsymbol{\Gamma}^\top \mathcal{C}(\mathbf{U})$, a subspace of dimension d (the map $\boldsymbol{\Lambda}^{-1/2} \boldsymbol{\Gamma}^\top$ is injective on $\mathcal{C}(\mathbf{V}) \supseteq \mathcal{C}(\mathbf{U})$). By the computation in (e), $\text{SSE}_{\mathbf{V}} = \|(\mathbf{I} - \mathbf{M}_d) \mathbf{Z}_*\|^2$ where $\mathbf{M}_d$ is the orthogonal projection of $\mathbb{R}^q$ onto that d -dimensional subspace, and $\mathbf{X}\tilde{\boldsymbol{\beta}}$ is a fixed vector plus a function of $\mathbf{M}_d \mathbf{Z}_*$, since $\mathbf{N}^\top \mathbf{Y}$ is constant by (a). Now Theorem 2.3.1 gives $E(\text{SSE}_{\mathbf{V}}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}_d) = \sigma^2(q - d)$ with no distributional assumption, and under normality Theorem 7.5.1 applied to $\mathbf{Z}_*$ gives the $\chi^2(q - d)$ law and the independence. Finally $\text{rank}(\mathbf{X} : \mathbf{V}) = r + q - d$ by the dimension formula for a sum of subspaces, so $q - d = \text{rank}(\mathbf{X} : \mathbf{V}) - r$.

■

The singular model is the nonsingular model with a deterministic layer on top. Part (b) isolates

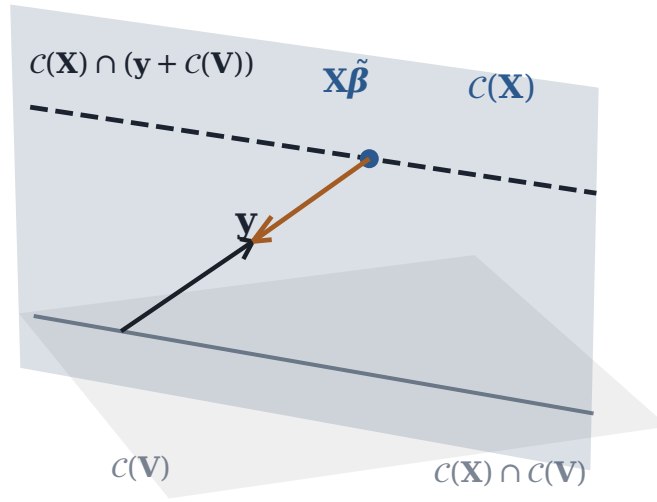


Figure 31.3.1: The geometry of a singular model with $n = 3$. The plane is $\mathcal{C}(\mathbf{X})$; $\mathcal{C}(\mathbf{V})$ is horizontal, so $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V})$ is the lower line. The data $\mathbf{y}$ determine $\mathbf{N}^\top \mathbf{X} \boldsymbol{\beta}$ exactly, which confines $\mathbf{X} \boldsymbol{\beta}$ to the dashed line $\mathcal{C}(\mathbf{X}) \cap (\mathbf{y} + \mathcal{C}(\mathbf{V}))$; the estimate is the point of that line nearest $\mathbf{y}$ in the $\mathbf{V}^+$ geometry, and the residual lies in $\mathcal{C}(\mathbf{V})$.

that layer, $r - d$ linear functions of $\boldsymbol{\beta}$ known exactly once the data are seen; part (e) pins down a particular solution with them and then does generalized least squares inside $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V})$, the part of the model space the noise can still reach; part (f) counts what is left. If $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{V})$ then $d = r$, nothing is known exactly, and Eq. 31.3.3 is generalized least squares with $\mathbf{V}^+$ for $\mathbf{V}^{-1}$ on $q - r$ degrees of freedom; if $\mathbf{V}$ is nonsingular then $q = n$, $d = r$, and everything collapses to Theorem 31.1.1.

31.3.4 A worked example

Example. Regression on estimated treatment effects

A balanced one-way layout with a treatments and m observations each is analysed, and the estimated effects $\hat{\alpha}_1, \dots, \hat{\alpha}_a$ are reported under the side condition $\sum_k \hat{\alpha}_k = 0$ (Theorem 8.4.2). A second analyst, with access only to these numbers, regresses them on the dose x_k applied to treatment k , fitting $E(\hat{\alpha}_k) = \beta_0 + \beta_1 x_k$ with an intercept.

The reported vector has $\text{Cov}(\hat{\boldsymbol{\alpha}}) = \tau^2 \mathbf{V}$ with $\mathbf{V} = \mathbf{I}_a - \mathbf{1}\mathbf{1}^\top/a$ and $\tau^2 = \sigma^2/m$, because $\hat{\alpha}_k = \bar{Y}_k - \bar{Y}$ and the $\bar{Y}_k$ are uncorrelated with variance τ^2 . Here $q = a - 1$, $\mathcal{C}(\mathbf{V}) = \mathbf{1}^\perp$, and $\mathbf{N} = \mathbf{1}/\sqrt{a}$. With $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ we get $d = 1$ and $r = 2$, so Theorem 31.3.2(b) gives exactly $r - d = 1$ perfectly known function: $\mathbf{X}^\top \mathbf{N} \propto (a, \sum_k x_k)^\top$, so

$$a\beta_0 + \left(\sum_k x_k\right)\beta_1 = \mathbf{1}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{1}^\top \hat{\boldsymbol{\alpha}} = 0$$

with probability one. In words: the intercept is determined by the slope, $\beta_0 = -\bar{x}\beta_1$, and the data say so without error. Fitting an intercept was a mistake, and the model itself reveals it.

Everything else is unchanged. Because $\mathbf{V}\mathbf{X} = \mathbf{X} - \mathbf{1}(\mathbf{1}^\top \mathbf{X})/a$ has columns in $\mathcal{C}(\mathbf{X})$, Theorem 31.2.2 applies and ordinary least squares is already best: $\hat{\beta}_1$ is the usual centred slope and $\hat{\beta}_0 = -\bar{x}\hat{\beta}_1$. Since $\mathbf{V}$ is idempotent, $\mathbf{V}^+ = \mathbf{V}$ and $\text{SSE}_{\mathbf{V}} = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}}$ is the usual residual sum of squares, on $q - d = a - 2$ degrees of freedom. What is *not* unchanged is the covariance matrix. Because $\mathbf{1}^\top \hat{\boldsymbol{\alpha}} = 0$ exactly,

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \frac{\tau^2}{S_{xx}}, & \text{Var}(\hat{\beta}_0) &= \bar{x}^2 \text{Var}(\hat{\beta}_1), \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= -\bar{x} \text{Var}(\hat{\beta}_1),\end{aligned}$$

whereas the ordinary formula $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ reports $s^2(1/a + \bar{x}^2/S_{xx})$ for the intercept. With $a = 5$ doses 0, 1, 2, 4, 8, $m = 6$ replicates and $\sigma = 1.2$, so $\tau^2 = 0.24$, $\bar{x} = 3.0$ and $S_{xx} = 40.0$: the true variances are 0.00600 for the slope and 0.0540 for the intercept, while the ordinary formula reports 0.1020 on average for the intercept, too large by a factor of 1.89. A simulation of 200,000 layouts reproduces 0.00600 and 0.0540.

```
import numpy as np

dose = np.array([0.0, 1.0, 2.0, 4.0, 8.0])           # a = 5 treatments
a, reps, sigma = len(dose), 6, 1.2
X = np.column_stack([np.ones(a), dose])
V = np.eye(a) - np.ones((a, a)) / a                # Cov(effects) = (sigma^2/reps)
V
xbar = dose.mean()
Sxx = float(((dose - xbar) ** 2).sum())

Vplus = V                                           # V is symmetric idempotent, so
V+ = V
N = np.ones((a, 1)) / np.sqrt(a)                  # an orthonormal basis of C(V)
perp
print("rank(V) =", np.linalg.matrix_rank(V), " rank([X V]) =",
      np.linalg.matrix_rank(np.column_stack([X, V])), " rank(X) =", np.linalg.
      matrix_rank(X))
print("the perfectly known function:", (X.T @ N).ravel())
```

[Run this code online](#)

```
rng = np.random.default_rng(3107)
trials = 200_000
cell = rng.normal(scale=sigma / np.sqrt(reps), size=(trials, a))   # treatment
means
cell += 0.35 * (dose - xbar)                                         # the true
effects
effects = cell - cell.mean(axis=1, keepdims=True)                  # sum to zero
exactly
B = effects @ np.linalg.solve(X.T @ X, X.T).T                      # ordinary
least squares

tau2 = sigma**2 / reps
print("Var(slope):      %.5f simulated, %.5f from tau^2/Sxx" % (B[:, 1].var(), tau2 /
      Sxx))
print("Var(intercept): %.5f simulated, %.5f from xbar^2 tau^2/Sxx"
```

```
% (B[:, 0].var(), xbar**2 * tau2 / Sxx))
print("what OLS reports for the intercept: %.5f" % (tau2 * (1 / a + xbar**2 / Sxx)))
```

Run this code online

The script behind [Example \(Regression on estimated treatment effects\)](#) also computes the estimate four ways — by [Eq. 31.3.2](#), by the closed form [Eq. 31.3.3](#), by [Eq. 31.3.1](#) with $\mathbf{T} = \mathbf{V} + \mathbf{X}\mathbf{X}^\top$, and by ordinary least squares — and checks that all four agree, as [Theorem 31.3.2\(d\)](#) requires, and that [Eq. 31.3.3](#) gives the same answer for two different generalized inverses $\mathbf{K}^-$.

Warning. Software that inverts $\mathbf{V}$ will fail here, and software that silently replaces $\mathbf{V}^{-1}$ by a pseudo-inverse solves the wrong problem: it minimizes the $\mathbf{V}^+$ criterion over all $\mathbf{b}$ instead of over $\mathcal{F}(\mathbf{y})$, which is what [Theorem 31.3.2\(e\)](#) is for, so the point estimate can be wrong. In [Example \(Regression on estimated treatment effects\)](#) the formula $(\mathbf{X}^\top \mathbf{V}^+ \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^+ \mathbf{y}$ always returns $\hat{\beta}_0 = 0$, contradicting $\mathbf{1}^\top \hat{\boldsymbol{\alpha}} = 0$. The smallest instance has two observations, the second measured without error: $\mathbf{V} = \text{diag}(1, 0)$, $\mathbf{X} = \mathbf{1}_2$, so $\beta = Y_2$ exactly, while the formula reports Y_1 . The degrees of freedom are wrong too: $n - r$ is not $q - d$. Here $a - 2 = 3$ happens to agree with $n - r$, but with a $\mathbf{V}$ of lower rank it would not; always compute $\text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$.

31.3.5 Testing

A reduced model $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ is tested as in [Theorem 31.1.1\(e\)](#), with two adjustments. The degrees of freedom are those of [Theorem 31.3.2\(f\)](#) for each model, $\nu = \text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$ and $\nu_0 = \text{rank}(\mathbf{X}_0 : \mathbf{V}) - \text{rank}(\mathbf{X}_0)$, so the numerator carries $\nu_0 - \nu$. And the reduced model can be *refuted with certainty*: if $\mathbf{Y} \notin \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$ it is impossible and the test rejects at once. Otherwise [Eq. 31.1.7](#) has the $F(\nu_0 - \nu, \nu, \gamma)$ distribution; [Exercise \(C1\)](#) works through the argument.

31.3.6 Exercises

31.3.6.1 A. Check your understanding

Exercise. A1

Show that if $\mathbf{V}$ is singular with $\mathbf{V}\mathbf{a} = \mathbf{0}$, then $\mathbf{a}^\top \mathbf{Y}$ is the same number for every possible data vector, and identify that number in terms of $\boldsymbol{\beta}$. What does this say about $\mathbf{a}^\top \mathbf{X}\boldsymbol{\beta}$ when $\mathbf{X}^\top \mathbf{a} = \mathbf{0}$?

Exercise. A2

Verify $\text{rank}(\mathbf{X} : \mathbf{V}) = \text{rank}(\mathbf{X}) + \text{rank}(\mathbf{V}) - \dim(\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V}))$ and compute both sides for [Example \(Regression on estimated treatment effects\)](#).

Solution. This is the dimension formula for the sum of two subspaces, $\dim(\mathcal{S} + \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T} - \dim(\mathcal{S} \cap \mathcal{T})$. In [Example \(Regression on estimated treatment effects\)](#), $\mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V}) = \mathbb{R}^a$ because $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathcal{C}(\mathbf{V}) = \mathbf{1}^\perp$; so the left side is a , and the right side is $2 + (a - 1) - 1 = a$.

31.3.6.2 B. Practice

Exercise. B1

Three fractions of a sample are measured and then rescaled so that they add to one. Model $E(Y_k) = \mu + \delta_k$ with $\sum_k \delta_k = 0$ known, and suppose $\text{Cov}(\mathbf{Y}) = \sigma^2(\mathbf{I}_3 - \mathbf{1}\mathbf{1}^\top/3)$. Which function of $(\mu, \delta_1, \delta_2)$ is known exactly? Find the variance of $Y_1 - Y_2$ as an estimate of $\delta_1 - \delta_2$, and explain why no other linear unbiased estimate exists.

Solution. Here $\mathbf{N} = \mathbf{1}/\sqrt{3}$ and $\mathbf{X}^\top \mathbf{N} \propto \mathbf{X}^\top \mathbf{1}$; with the parameterization $\boldsymbol{\beta} = (\mu, \delta_1, \delta_2)^\top$ and $\delta_3 = -\delta_1 - \delta_2$, we get $\mathbf{X}^\top \mathbf{1} = (3, 0, 0)^\top$, so 3μ is known exactly: $\mu = \mathbf{1}^\top \mathbf{Y}/3 = 1/3$, the rescaling constant. Here $\text{rank}(\mathbf{X}) = 3 = n$, so $\mathbf{X}^\top$ is nonsingular and the unbiasedness equation $\mathbf{X}^\top \mathbf{c} = \boldsymbol{\lambda}$ has exactly one solution: every estimable function has a single linear unbiased estimate, which is therefore its BLUE. For $\delta_1 - \delta_2$ that solution is $\mathbf{c} = (1, -1, 0)^\top$, and $\text{Var}(Y_1 - Y_2) = \sigma^2 \mathbf{c}^\top \mathbf{V} \mathbf{c} = \sigma^2(2/3 + 2/3 + 2/3) = 2\sigma^2$.

Exercise. B2

Suppose the first observation is measured without error, so $\mathbf{V} = \begin{pmatrix} 0 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{V}_{22} \end{pmatrix}$ with $\mathbf{V}_{22}$ positive definite, and $\mathbf{X} = \begin{pmatrix} \mathbf{x}_{(1)}^\top \\ \mathbf{X}_2 \end{pmatrix}$. Show that $\mathbf{x}_{(1)}^\top \boldsymbol{\beta}$ is known exactly, and that the BLUE of any estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is the generalized least squares estimate from the remaining $n-1$ observations subject to the constraint $\mathbf{x}_{(1)}^\top \mathbf{b} = y_1$.

Solution. $\mathbf{N} = \mathbf{e}_1$ spans $\mathcal{C}(\mathbf{V})^\perp$, so $\mathbf{N}^\top \mathbf{Y} = Y_1 = \mathbf{x}_{(1)}^\top \boldsymbol{\beta}$ with probability one. In Eq. 31.3.2 the feasible set is $\{\mathbf{b} : \mathbf{y} - \mathbf{X}\mathbf{b} \in \mathcal{C}(\mathbf{V})\} = \{\mathbf{b} : \mathbf{x}_{(1)}^\top \mathbf{b} = y_1\}$, and on $\mathcal{C}(\mathbf{V})$ the form $\mathbf{a}^\top \mathbf{V}^+ \mathbf{a}$ is $\mathbf{a}_2^\top \mathbf{V}_{22}^{-1} \mathbf{a}_2$; so the criterion is the generalized least squares criterion of the last $n-1$ observations, minimized subject to the constraint. This is restricted least squares (Theorem 9.2.5) in the $\mathbf{V}_{22}^{-1}$ geometry.

Exercise. B3

Verify directly from Eq. 31.3.3 that $\mathbf{X}\tilde{\boldsymbol{\beta}}$ does not depend on which generalized inverse $\mathbf{K}^-$ is used, given $\mathbf{y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})$.

31.3.6.3 C. Going deeper

Exercise. C1

In the setting of Theorem 31.3.2, let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with $\nu_0 = \text{rank}(\mathbf{X}_0 : \mathbf{V}) - \text{rank}(\mathbf{X}_0)$ and $\nu = \text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$, and assume $\mathbf{Y}$ is normal and $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$. Working in the coordinates $\mathbf{Z}_*$ of the proof of Theorem 31.3.2(f), show that $\text{SSE}_{\mathbf{V},0} - \text{SSE}_{\mathbf{V}}$ and $\text{SSE}_{\mathbf{V}}$

are independent with $(\text{SSE}_{\mathbf{V},0} - \text{SSE}_{\mathbf{V}})/\sigma^2 \sim \chi^2(\nu_0 - \nu, \gamma)$, and identify γ . Why is the assumption $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$ needed?

Solution. Because $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$ makes $\mathbf{N}^\top \mathbf{Y} \in \mathcal{C}(\mathbf{N}^\top \mathbf{X}_0)$, the same $\mathbf{b}_*$ is feasible in both models. In the coordinates $\mathbf{Z}_* \sim N_q(\boldsymbol{\mu}_*, \sigma^2 \mathbf{I})$ the two sums of squares are $\|(\mathbf{I} - \mathbf{M}_d)\mathbf{Z}_*\|^2$ and $\|(\mathbf{I} - \mathbf{M}_{d_0})\mathbf{Z}_*\|^2$ for nested subspaces of dimensions $d_0 \leq d$, with $q-d = \nu$ and $q-d_0 = \nu_0$. Now Theorem 11.1.1 applied to $\mathbf{Z}_*$ gives independence, the χ^2 laws and $\gamma = \|(\mathbf{M}_d - \mathbf{M}_{d_0})\boldsymbol{\mu}_*\|^2/\sigma^2$. If $\mathbf{X}\boldsymbol{\beta} \notin \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$ the reduced model assigns probability zero to the observed $\mathbf{Y}$, so it is contradicted outright and no distribution theory is needed.

Exercise. C2

Call $\mathbf{X}\tilde{\boldsymbol{\beta}}$ a *least squares consistent estimate* when it minimizes the ordinary sum of squares $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ over $\mathcal{F}(\mathbf{y})$; this is the criterion used by Christensen (2020). Show that it agrees with Eq. 31.3.2 when $\mathbf{V}$ is idempotent, and that in general the two differ. Show also that when $\mathcal{C}(\mathbf{V}\mathbf{U}) \subseteq \mathcal{C}(\mathbf{U})$ the least squares consistent estimate is a BLUE.